

SENTINEL AI

**Detect threats. Predict attacks.
Respond intelligently.**

A modular cyber threat intelligence and SOC platform that turns security telemetry into explainable decisions.



Security teams see signals. They need decisions.

Authentication events, alerts, threat intelligence, and incidents are often separated by tools and time.

01 / FRAGMENTED

Context breaks

Logs, identities, IP reputation, detections, and response evidence live in disconnected systems.

02 / OVERLOADED

Noise wins

High-volume alerts lack prioritization, explainability, and a visible connection to business risk.

03 / DELAYED

Response waits

Analysts spend time reconstructing evidence before they can contain or escalate a threat.

THE GAP

A continuous, explainable loop—from security event to controlled response.

One defensive pipeline. One operational view.

Sentinel AI preserves identity, source, risk, and evidence as each signal moves through the platform.



THE SENTINEL VALUE

Faster understanding • Explainable risk • Predictive context • Auditable response

A single security event remains traceable across logs, alerts, detections, paths, incidents, notifications, and reports.

A threat story judges can see end-to-end

The current implementation turns repeated authentication failures into visible, persisted security outcomes.



LOGIN

Success and failure events capture user, IP, user agent, status, and time.



DETECT

>5 failures for the same username and IP inside five minutes.



SCORE

A transparent HIGH / 90 risk decision with behavior evidence.



RESPOND

Alert, threat, attack path, incident, notification, and report become visible.



IMPLEMENTED + VERIFIED

Built for the analyst's next decision

The interface unifies monitoring, intelligence, detection, prediction, response, and reporting.

CAPABILITY

LIVE MONITORING

Authentication timeline, filters, status, device and source context.

CAPABILITY

THREAT INTELLIGENCE

Suspicious IPs, threat types, risk levels, and enrichment.

CAPABILITY

AI DETECTION

Risk score, confidence, and explainable behavior analysis.

CAPABILITY

ATTACK GRAPH

Possible progression from entry point to sensitive asset.

CAPABILITY

SOAR RESPONSE

Incident workflow, notifications, and defensive action state.

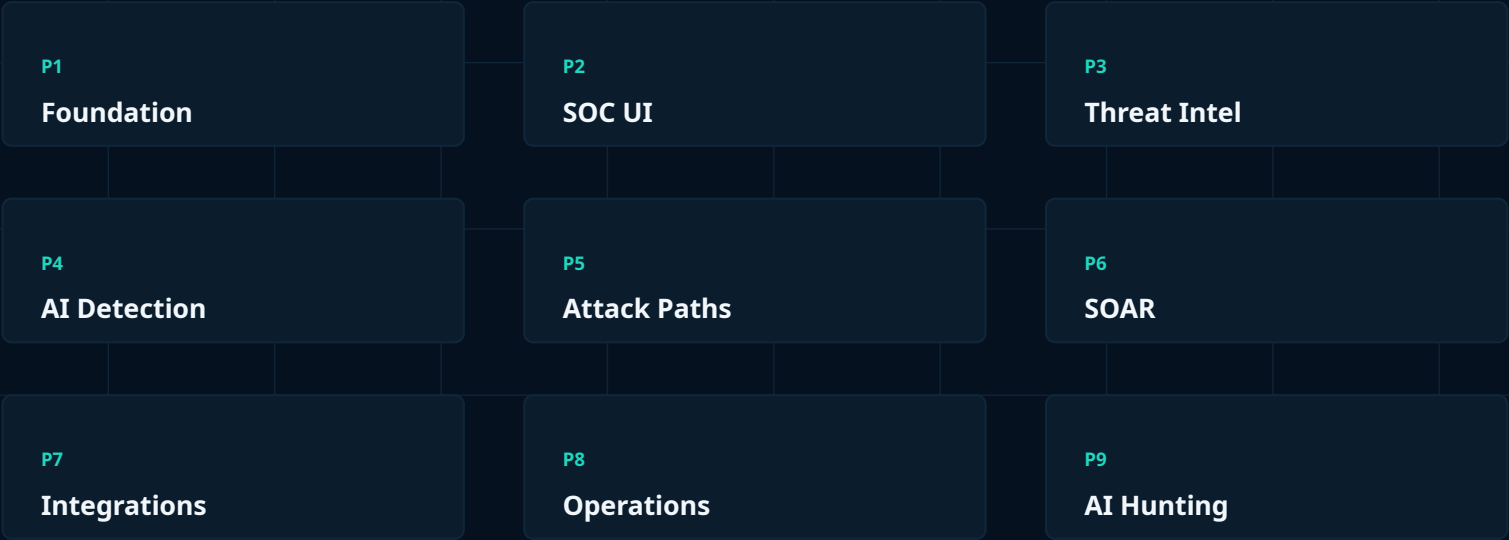
CAPABILITY

REPORTING

Live summary, archives, JSON and CSV exports.

Modular by design. Integrated by a central security plane.

Nine independently runnable phases keep domain ownership while the central API coordinates identity, data, and operations.



CENTRAL SECURITY PLANE

FastAPI Gateway

Authentication + APIs
Detection orchestration
Health + topology
Fault-tolerant aggregation
PostgreSQL persistence

REACT

POSTGRESQL

DOCKER

The platform protects its own trust boundary

Production-minded controls are implemented around identity, secrets, persistence, and defensive automation.

CONTROL

PBKDF2

600,000 password-hashing iterations with a per-user salt.

CONTROL

TOKEN SAFETY

Random bearer tokens; only SHA-256 digests persist.

CONTROL

EXPIRY + LOGOUT

Expiring sessions and explicit server-side revocation.

CONTROL

SECRET HYGIENE

Credentials supplied through environment configuration.

CONTROL

SAFE MIGRATION

Additive initialization preserves existing users and data.

CONTROL

AUDITABILITY

Success, failure, IP, user agent, reason, and time recorded.

Explainable detection before opaque automation

Deterministic rules provide a testable baseline; AI modules extend analysis without hiding the evidence.

10 / 100

NORMAL ACTIVITY

35-55

SUSPICIOUS PATTERN

90 / 100

BRUTE FORCE

WHY THIS SCORE?

Every decision keeps its evidence trail.

- username and identity
- source IP activity
- attempt frequency
- five-minute window
- distinct usernames
- failure reason

CURRENT VERIFIED RULE: same username + same IP + more than five failed attempts + five-minute window.

Honest coverage today. Extensible coverage tomorrow.

Sentinel AI does not claim to detect every attack; it provides a modular path to broader evidence-based detection.

IMPLEMENTED + VERIFIED

Authentication threats

- brute-force login
- repeated failed attempts
- suspicious login frequency
- multiple usernames from one IP
- identity/IP risk scoring
- active-alert deduplication
- threat and incident promotion

ROADMAP MODULES

WEB

SQL injection
XSS
command injection
path traversal
SSRF
malicious upload

IDENTITY

credential stuffing
password spraying
session anomalies
impossible travel
new device/location

NETWORK + ENDPOINT

port scanning
DDoS indicators
DNS tunneling
lateral movement
ransomware behavior
C2 activity

ROADMAP ≠ CURRENT CLAIM

A safe blueprint for web-attack detection

Sanitized server/WAF telemetry can extend the existing event pipeline without storing secrets or raw credentials.



SQL INJECTION

Proposed signals

SQL operator anomalies, UNION patterns, comment markers, encoded payload attempts, database error bursts, abnormal parameter lengths.

CROSS-SITE SCRIPTING

Proposed signals

Script/event-handler patterns, encoded markup, repeated payload families, CSP/WAF evidence, and response anomalies.

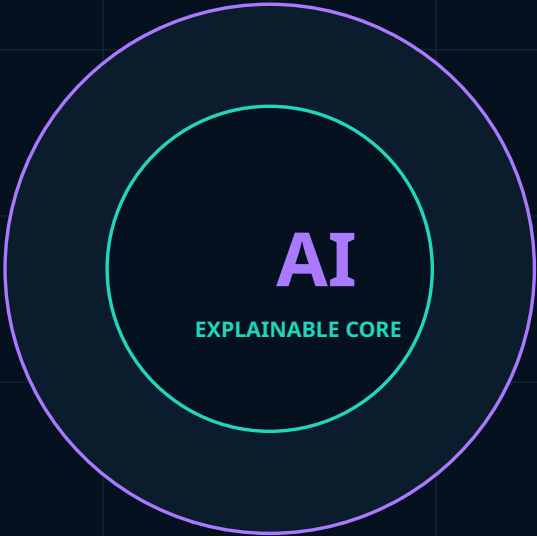
OTHER WEB THREATS

Proposed signals

Traversal, command injection, SSRF, file inclusion, malicious upload, API abuse, XML-RPC abuse, and web-shell indicators.

AI augments evidence. It does not replace it.

Phase 4 extends transparent rules with bounded anomaly analysis, classification, and confidence scoring.



AI STAGE

FEATURES

frequency • identity • IP • context

AI STAGE

ANOMALY

deviation from known behavior

AI STAGE

CLASSIFY

threat type + bounded severity

AI STAGE

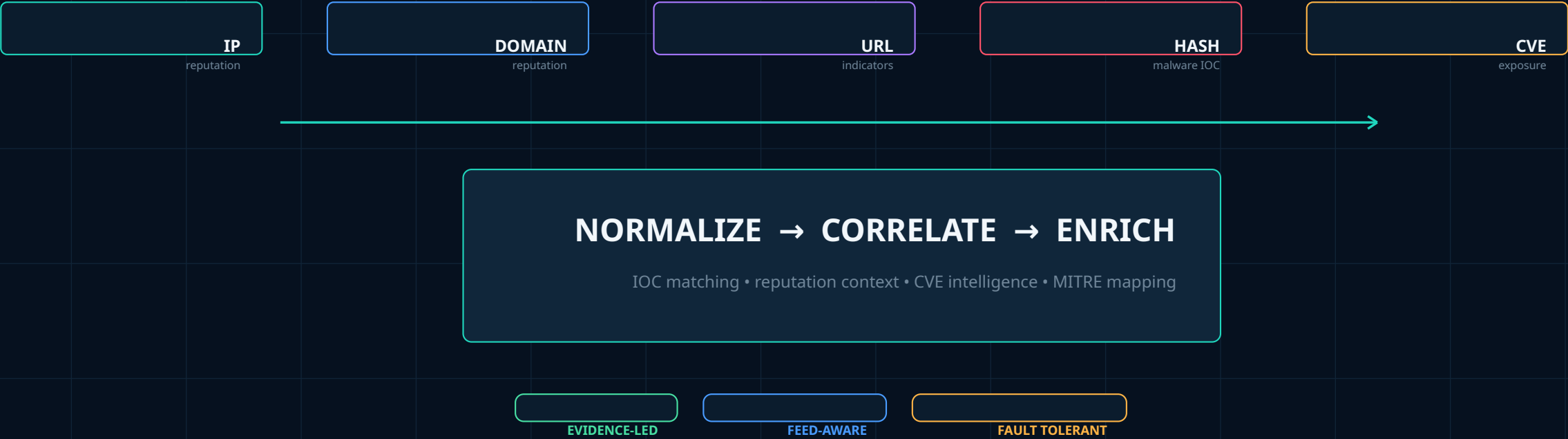
EXPLAIN

human-readable reason + evidence

Human-readable reasoning remains visible beside every score.

Detection becomes stronger when context travels with it

Sentinel AI's modular intelligence services normalize external evidence and correlate it with internal events.



Show what could happen next—not only what happened

Attack-path context helps analysts connect an entry signal to possible privilege and asset exposure.



ANALYST VALUE

Source identity • Possible progression • Risk propagation • Defensive recommendation

Prediction is presented as bounded context—not certainty—and remains linked to the originating evidence.

Automate the workflow. Keep control of the action.

Sentinel AI records, prioritizes, and coordinates defensive response without blindly executing disruptive changes.

WORKFLOW

CREATE

incident

WORKFLOW

PRIORITIZE

severity

WORKFLOW

NOTIFY

administrator

WORKFLOW

PREPARE

containment

WORKFLOW

APPROVE

disruptive action

WORKFLOW

AUDIT

outcome

HUMAN APPROVAL BOUNDARY

Account locks, IP blocks, and other business-impacting actions require authorization.

Security posture that is measurable and exportable

Live summaries and generated reports reuse existing PostgreSQL evidence—without duplicate security tables.

LOGIN

TOTAL ATTEMPTS

SUCCESS

AUTHENTICATED

FAILED

DENIED

ALERTS

ACTIVE

RISK

LOW-HIGH

REPORT CENTER

Daily Security • Threat Analysis • Incident Report • Archive

Authenticated API → PostgreSQL evidence → live cards → JSON / CSV exports

JSON

CSV

AUDITABLE

Validated as a working system—not a static concept

The project includes executable services, persistence, protected APIs, a connected frontend, and repeatable tests.

VERIFIED

BUILD

React production bundle

VERIFIED

COMPILE

FastAPI modules

VERIFIED

TEST

report routes + summary

VERIFIED

RUN

Docker + PostgreSQL

VERIFIED

VERIFY

authenticated 200 API

DEMO SUCCESS CRITERIA

Failed login → six attempts → HIGH alert → risk → attack path → notification → report

No invented customer metrics. No unsupported accuracy claims. Every claim maps to code, API, database, or roadmap.

A deliberate path from working prototype to broader defense

Expand evidence sources and detections without rewriting the central architecture.

NOW

HARDEN

RBAC • streaming • tests • observability •
deployment

NEXT

EXPAND WEB

WordPress telemetry • SQLi • XSS • traversal
• API abuse

SCALE

ENTERPRISE

IDS/EDR • multi-tenancy • HA • compliance •
collaboration



Turn every security signal into a confident response.

Sentinel AI connects detection, understanding, prediction, and controlled response in one explainable SOC workflow.

